



SAHAYAK

Offline. Vernacular. On your side.

A scam shield and benefits guide that runs on one node at a CSC counter, with no internet, in Hindi and English, by voice or touch. Built and measured.

Ideas for India 2026 · Inclusive Innovation for Bharat

Roshan Raj · Ishmiit Singh

Banked, then left to be robbed

Hundreds of millions of first-time account holders now receive OTPs, UPI requests and scheme messages they were never taught to read.

Fraud preys on the new

Fake KYC SMS, "scan to receive" QR codes, digital-arrest calls, fake scheme apps: built to fool people who cannot yet tell a real bank message from a trap.

Benefits go unclaimed

Pensions, insurance and income support are lost to paperwork and to not knowing they exist, or which office to ask.

Tools assume the connected

Every safety tool assumes internet, a smartphone, English and the cloud: the four things the excluded lack.

One offline node. Two questions answered.

Phones join the node's Wi-Fi, which has no internet, and ask in Hindi or English, by voice or touch.

"Is this a scam?"

Paste, speak, photograph or describe a message, a call or a UPI QR. Verdict in milliseconds, the three reasons, what to do, and a ready 1930 complaint.

"What am I owed?"

A spoken interview (median 5 questions) across 12 central schemes: what you can get, the papers to carry, where to go, what to say, and a printed slip.

A counter, not an app

The CSC or bank agent sees "Ask the agent" requests, prints 58 mm slips and reads monthly impact counters that hold no personal data.

Built, and on a phone today



A fake KYC SMS



Widow, 67: what she can claim



"Scan to get cashback" QR

Technology and implementation

01

Phone app

A cached web app (no install), Hindi and English, every screen read aloud; works on a basic Android phone.

02

Fraud-Shield

51 named signals in a signed content pack, a pattern matcher and a small classifier decide. Vetted templates explain; an optional local LLM may add a line, only through a safety gate.

03

Benefits Navigator

Scheme rules as data, each traced to its official page and date; yes / likely / unknown / no logic; asks only what can change the answer.

04

Offline voice

Speech in and out on the node (Vosk, Piper); amounts spoken as words; "I heard ..., is that right?" before any answer is used.

05

The node

One laptop or mini-PC at a CSC. Zero-egress monitor, Ed25519-signed packs updated by USB, operator console, captive portal.

What no one else ships together

Offline, and it proves it

The node counts every outbound connection its own process makes, live on screen, next to the firewall's state. The judge's phone shows "no internet".

Explains, not a blocklist

Named signals a judge can check. On ScamBench, 91.9% of scams caught against 51.4% for a keyword blocklist, with 8.7% false alarms against 43.5%.

Rules, not guesses

Scheme answers computed from official rules: 2,400 of 2,400 decisions match an independent re-derivation; every rupee amount traces to a signed pack.

It already works, and we measured it

91.9%

scams caught (blocklist: 51.4%)

2 ms

per verdict on a laptop CPU

15.4%

Hindi word error, real speakers

0.65 s

from mic release to spoken reply

5

questions to find a person's schemes

How we know, and what it does not show

ScamBench v0 frozen test split (60 messages, 95% CI 82.1%–100.0%); SchemeBench (200 personas, exhaustive interview check); Google FLEURS Hindi (418 recordings). v0 messages were written by our team, so those numbers are optimistic; v1 adds real messages collected with consent. Red team: 46 of 52 scams disguised to slip past it caught on the first run (blocklist 33). All in the testing report.

From "account opened" to "citizen safe"

Rupees at risk

The node adds up the money asked for in messages it flags, every month: never called "saved", always shown to the district without personal data.

Schemes found

People found eligible, by scheme, with the papers and the office; each printed slip lets the CSC operator finish the form at the counter.

Nothing leaves

Messages, voices and answers stay in the node's memory; the case log needs consent, holds no names and deletes itself after 30 days.

Pilot target, not a result: one node per CSC in a 20-village district, with an operator and real users. The counters are how the district will know.

Grow by replication, not reinvention

Replicable node

A deployment is a node image plus signed content packs plus a one-page operator guide. A new scam pattern or scheme is a pack update by USB, not a code change.

Rides existing rails

Delivered through CSCs, bank business correspondents and ASHA workers: networks people already trust, with an operator at the counter.

Sustainable by design

No cloud or per-query cost; open, testable rules; counters give the district a monthly number to fund against.

Built on India's own rails

Digital India, last mile

Extends digital public services to the citizens they reach last.

Jan Dhan · DBT · UPI

Protects the financial-inclusion base the nation has already built.

1930 and cybercrime.gov.in

Every scam verdict ends in the national helpline and a ready complaint.

Indian languages

Hindi and English by voice today; the language data is a pack, so more languages follow.

DPDP Act

Data stays on the node: consent first, nothing named, deleted by default.

CSC · BC · ASHA

Delivered through the public networks already trusted on the ground.

Builders who ship, seeking last-mile partners

Roshan Raj

Systems and AI-infrastructure engineer · Manipal Institute of Technology · 7 patents pending

Ishmiit Singh

Co-builder: product, content and evidence

Together: the node, the scam shield, the scheme navigator, offline voice, the operator console and the benchmarks behind every number in this deck.

The ask

A single-district pilot with the natural last-mile partners:

- CSC e-Governance / State IT: village access points
- A bank BC network / SLBC: reach and real messages for ScamBench v1
- A reviewer for scheme rules and money answers
- A rural NGO: field testing and Hindi review



India banked half a billion people.

Sahayak helps keep them safe.

Offline · vernacular · privacy-first · built and measured. The connectivity divide, turned into a bridge.

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